

WEEK -4

PRACTICAL

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Sub : - DBSE&DBD

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Microsoft Windows [Version 10.0.26200.8875]
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C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE db_2;
Query OK, 1 row affected (0.01 sec)

mysql> USE db_2;
Database changed
mysql> CREATE TABLE Departments (
->     dept_id INT PRIMARY KEY,
->     dept_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Employees (
->     emp_id INT PRIMARY KEY,
->     emp_name VARCHAR(50),
->     dept_id INT,
->     manager_id INT,
->     salary DECIMAL(10,2),
->     email VARCHAR(100)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Projects (
->     project_id INT PRIMARY KEY,
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mysql>
mysql> CREATE TABLE Projects (
->     project_id INT PRIMARY KEY,
->     project_name VARCHAR(100),
->     emp_id INT
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Customers (
->     customer_id INT PRIMARY KEY,
->     customer_name VARCHAR(50),
->     email VARCHAR(100)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Products (
->     product_id INT PRIMARY KEY,
->     product_name VARCHAR(100),
->     price DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Orders (
->     order_id INT PRIMARY KEY,
->     customer_id INT,
->     order_date DATE
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Order_Details (
->     order_id INT,
->     product_id INT,
->     quantity INT,
->     PRIMARY KEY (order_id, product_id)
-> );
Query OK, 0 rows affected (0.02 sec)
```

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mysql>
mysql> INSERT INTO Departments VALUES
-> (1, 'CSE'),
-> (2, 'ECE'),
-> (3, 'EEE'),
-> (4, 'MECH'),
-> (5, 'CIVIL');
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Employees VALUES
-> (1, 'Rahul', 1, NULL, 80000, 'rahul@gmail.com'),
-> (2, 'Arjun', 1, 1, 65000, 'arjun@gmail.com'),
-> (3, 'Priya', 1, 1, 55000, 'priya@gmail.com'),
-> (4, 'Sneha', 2, 1, 60000, 'sneha@gmail.com'),
-> (5, 'Kiran', 2, 1, 70000, 'kiran@gmail.com'),
-> (6, 'Anil', 3, 2, 28000, 'anil@gmail.com'),
-> (7, 'Vijay', 1, 1, 90000, 'vijay@gmail.com'),
-> (8, 'Meena', 3, 2, 50000, 'meena@gmail.com'),
-> (9, 'Ravi', 1, 1, 95000, 'ravi@gmail.com'),
-> (10, 'Pooja', 4, 5, 35000, 'pooja@gmail.com'),
-> (11, 'Suresh', NULL, 1, 58000, 'suresh@gmail.com'),
-> (12, 'Neha', 5, 5, 48000, 'neha@gmail.com'),
-> (13, 'Amit', 1, 1, 72000, 'amit@gmail.com'),
-> (14, 'Divya', 1, 1, 68000, 'divya@gmail.com'),
-> (15, 'Rohit', 2, 5, 30000, 'rohit@gmail.com'),
-> (16, 'Manoj', 3, 2, 25000, 'manoj@gmail.com');
Query OK, 16 rows affected (0.00 sec)
Records: 16  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Projects VALUES
-> (101, 'AI Project', 1),
-> (102, 'Web Project', 2),
-> (103, 'IoT Project', 4),
-> (104, 'Database Project', 7);
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

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mysql> INSERT INTO Projects VALUES
-> (101, 'AI Project', 1),
-> (102, 'Web Project', 2),
-> (103, 'IoT Project', 4),
-> (104, 'Database Project', 7);
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Customers VALUES
-> (1, 'Customer1', 'c1@gmail.com'),
-> (2, 'Customer2', 'c2@gmail.com'),
-> (3, 'Customer3', 'c3@gmail.com'),
-> (4, 'Customer4', 'c4@gmail.com'),
-> (5, 'Customer5', 'c5@gmail.com'),
-> (6, 'Customer6', 'c6@gmail.com');
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Products VALUES
-> (1, 'Laptop', 60000),
-> (2, 'Mobile', 30000),
-> (3, 'Keyboard', 2000),
-> (4, 'Mouse', 1000),
-> (5, 'Monitor', 15000);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Orders VALUES
-> (101, 1, '2026-01-10'),
-> (102, 2, '2026-01-15'),
-> (103, 1, '2026-02-05'),
-> (104, 3, '2026-02-20'),
-> (105, 4, '2026-03-10'),
-> (106, 2, '2026-03-15');
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
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mysql>
mysql> INSERT INTO Order_Details VALUES
-> (101, 1, 2),
-> (101, 3, 3),
-> (102, 2, 2),
-> (102, 4, 5),
-> (103, 1, 1),
-> (103, 5, 2),
-> (104, 3, 10),
-> (105, 2, 3),
-> (105, 4, 4),
-> (106, 1, 1);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Employees
-> VALUES (17, 'Varun', 2, 5, 50000, 'varun@gmail.com');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> UPDATE Employees
-> SET salary = salary * 1.10
-> WHERE dept_id = (
->     SELECT dept_id
->     FROM Departments
->     WHERE dept_name = 'CSE'
-> );
Query OK, 7 rows affected (0.00 sec)
Rows matched: 7  Changed: 7  Warnings: 0

mysql>
mysql> DELETE FROM Employees
-> WHERE salary < 30000;
Query OK, 2 rows affected (0.00 sec)

mysql>
mysql> SELECT *
-> FROM Employees;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | manager_id | salary | email |
+-----+-----+-----+-----+-----+-----+
| 17 | Varun | 2 | 5 | 50000 | varun@gmail.com |
+-----+-----+-----+-----+-----+-----+

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mysql>
mysql> SELECT *
-> FROM Employees;

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | manager_id | salary | email |
+-----+-----+-----+-----+-----+-----+
| 1 | Rahul | 1 | NULL | 88000.00 | rahul@gmail.com |
| 2 | Arjun | 1 | 1 | 71500.00 | arjun@gmail.com |
| 3 | Priya | 1 | 1 | 60500.00 | priya@gmail.com |
| 4 | Sneha | 2 | 1 | 60000.00 | sneha@gmail.com |
| 5 | Kiran | 2 | 1 | 70000.00 | kiran@gmail.com |
| 7 | Vijay | 1 | 1 | 99000.00 | vijay@gmail.com |
| 8 | Meena | 3 | 2 | 50000.00 | meena@gmail.com |
| 9 | Ravi | 1 | 1 | 104500.00 | ravi@gmail.com |
| 10 | Pooja | 4 | 5 | 35000.00 | pooja@gmail.com |
| 11 | Suresh | NULL | 1 | 58000.00 | suresh@gmail.com |
| 12 | Neha | 5 | 5 | 48000.00 | neha@gmail.com |
| 13 | Amit | 1 | 1 | 79200.00 | amit@gmail.com |
| 14 | Divya | 1 | 1 | 74800.00 | divya@gmail.com |
| 15 | Rohit | 2 | 5 | 30000.00 | rohit@gmail.com |
| 17 | Varun | 2 | 5 | 50000.00 | varun@gmail.com |
+-----+-----+-----+-----+-----+-----+

15 rows in set (0.00 sec)

mysql>
mysql> SELECT e.emp_name, d.dept_name
-> FROM Employees e
-> JOIN Departments d
-> ON e.dept_id = d.dept_id;

+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Rahul | CSE |
| Arjun | CSE |
| Priya | CSE |
| Sneha | ECE |
| Kiran | ECE |
| Vijay | CSE |
| Meena | EEE |
| Ravi | CSE |
| Pooja | MECH |
| Neha | CIVIL |
| Amit | CSE |
| Divya | CSE |
+-----+-----+
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mysql>
mysql> SELECT e.emp_name, d.dept_name
-> FROM Employees e
-> JOIN Departments d
-> ON e.dept_id = d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Rahul    | CSE       |
| Arjun    | CSE       |
| Priya    | CSE       |
| Sneha    | ECE       |
| Kiran    | ECE       |
| Vijay    | CSE       |
| Meena    | EEE       |
| Ravi     | CSE       |
| Pooja    | MECH      |
| Neha     | CIVIL     |
| Amit     | CSE       |
| Divya    | CSE       |
| Rohit    | ECE       |
| Varun    | ECE       |
+-----+-----+
14 rows in set (0.00 sec)

mysql>
mysql> SELECT e.emp_name
-> FROM Employees e
-> LEFT JOIN Departments d
-> ON e.dept_id = d.dept_id
-> WHERE d.dept_id IS NULL;
+-----+
| emp_name |
+-----+
| Suresh   |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT d.dept_name, e.emp_name
-> FROM Departments d
-> LEFT JOIN Employees e
-> ON d.dept_id = e.dept_id;
+-----+-----+
```

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mysql>
mysql> SELECT d.dept_name, e.emp_name
-> FROM Departments d
-> LEFT JOIN Employees e
-> ON d.dept_id = e.dept_id;
+-----+-----+
| dept_name | emp_name |
+-----+-----+
| CSE       | Divya    |
| CSE       | Amit     |
| CSE       | Ravi     |
| CSE       | Vijay    |
| CSE       | Priya    |
| CSE       | Arjun    |
| CSE       | Rahul    |
| ECE       | Varun    |
| ECE       | Rohit    |
| ECE       | Kiran    |
| ECE       | Sneha    |
| EEE       | Meena    |
| MECH      | Pooja    |
| CIVIL     | Neha     |
+-----+-----+
14 rows in set (0.00 sec)

mysql>
mysql> SELECT e.emp_name AS Employee, m.emp_name AS Manager
-> FROM Employees e
-> LEFT JOIN Employees m
-> ON e.manager_id = m.emp_id;
+-----+-----+
| Employee | Manager |
+-----+-----+
| Rahul    | NULL    |
| Arjun    | Rahul   |
| Priya    | Rahul   |
| Sneha    | Rahul   |
| Kiran    | Rahul   |
| Vijay    | Rahul   |
| Meena    | Arjun   |
| Ravi     | Rahul   |
| Pooja    | Kiran   |
| Suresh   | Rahul   |
| Neha     | Kiran   |
| Amit     | Rahul   |
| Divya    | Rahul   |
| Rohit    | Kiran   |
| Varun    | Kiran   |
+-----+-----+
15 rows in set (0.00 sec)
```

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mysql>
mysql> SELECT COUNT(*) AS Total_Employees
-> FROM Employees;
+-----+
| Total_Employees |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT d.dept_name, AVG(e.salary) AS Average_Salary
-> FROM Departments d
-> JOIN Employees e
-> ON d.dept_id = e.dept_id
-> GROUP BY d.dept_id, d.dept_name;
+-----+-----+
| dept_name | Average_Salary |
+-----+-----+
| CSE       | 82500.000000   |
| ECE       | 52500.000000   |
| EEE       | 50000.000000   |
| MECH      | 35000.000000   |
| CIVIL     | 48000.000000   |
+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT d.dept_name, AVG(e.salary) AS Average_Salary
-> FROM Departments d
-> JOIN Employees e
-> ON d.dept_id = e.dept_id
-> GROUP BY d.dept_id, d.dept_name
-> HAVING AVG(e.salary) > 60000;
+-----+-----+
| dept_name | Average_Salary |
+-----+-----+
| CSE       | 82500.000000   |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT MAX(salary) AS Highest_Salary,
-> MIN(salary) AS Lowest_Salary
-> FROM Employees;
+-----+-----+
| Highest_Salary | Lowest_Salary |
+-----+-----+
|      104500.00 |       30000.00 |
+-----+-----+
```

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C:\Windows\System32\cmd.e X + v
mysql>
mysql> SELECT MAX(salary) AS Highest_Salary,
-> MIN(salary) AS Lowest_Salary
-> FROM Employees;
+-----+-----+
| Highest_Salary | Lowest_Salary |
+-----+-----+
| 104500.00 | 30000.00 |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT emp_name, salary
-> FROM Employees
-> WHERE salary > (
-> SELECT AVG(salary)
-> FROM Employees
-> );
+-----+-----+
| emp_name | salary |
+-----+-----+
| Rahul | 88000.00 |
| Arjun | 71500.00 |
| Kiran | 70000.00 |
| Vijay | 99000.00 |
| Ravi | 104500.00 |
| Amit | 79200.00 |
| Divya | 74800.00 |
+-----+-----+
7 rows in set (0.00 sec)

mysql>
mysql> SELECT emp_name, dept_id
-> FROM Employees
-> WHERE dept_id = (
-> SELECT dept_id
-> FROM Employees
-> WHERE emp_name = 'Rahul'
-> );
+-----+-----+
| emp_name | dept_id |
+-----+-----+
| Rahul | 1 |
| Arjun | 1 |
| Priya | 1 |
| Vijay | 1 |
| Ravi | 1 |
| Amit | 1 |
| Divya | 1 |
+-----+-----+
```

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mysql>
mysql> SELECT emp_name, salary
      -> FROM Employees
      -> WHERE salary = (
      ->     SELECT MAX(salary)
      ->     FROM Employees
      -> );
+-----+-----+
| emp_name | salary |
+-----+-----+
| Ravi     | 104500.00 |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT d.dept_name, COUNT(e.emp_id) AS Employee_Count
      -> FROM Departments d
      -> JOIN Employees e
      -> ON d.dept_id = e.dept_id
      -> GROUP BY d.dept_id, d.dept_name
      -> HAVING COUNT(e.emp_id) > 5;
+-----+-----+
| dept_name | Employee_Count |
+-----+-----+
| CSE       | 7 |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT emp_name, salary
      -> FROM (
      ->     SELECT emp_name,
      ->            salary,
      ->            DENSE_RANK() OVER (ORDER BY salary DESC) AS salary_rank
      ->     FROM Employees
      -> ) AS x
      -> WHERE salary_rank = 2;
+-----+-----+
| emp_name | salary |
+-----+-----+
| Vijay    | 99000.00 |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT emp_name, salary, dept_id
      -> FROM (
      ->     SELECT emp_name,
      ->            salary,
```

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mysql>
mysql> SELECT emp_name, salary, dept_id
      -> FROM (
      ->     SELECT emp_name,
      ->            salary,
      ->            dept_id,
      ->            AVG(salary) OVER (PARTITION BY dept_id) AS dept_avg
      ->     FROM Employees
      -> ) AS x
      -> WHERE salary > dept_avg;
+-----+-----+-----+
| emp_name | salary | dept_id |
+-----+-----+-----+
| Rahul    | 88000.00 | 1 |
| Vijay    | 99000.00 | 1 |
| Ravi     | 104500.00 | 1 |
| Sneha    | 60000.00 | 2 |
| Kiran    | 70000.00 | 2 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT c.customer_id, c.customer_name
      -> FROM Customers c
      -> LEFT JOIN Orders o
      -> ON c.customer_id = o.customer_id
      -> WHERE o.order_id IS NULL;
+-----+-----+
| customer_id | customer_name |
+-----+-----+
| 5 | Customer5 |
| 6 | Customer6 |
+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT p.product_name,
      ->            SUM(od.quantity) AS Total_Sold
      -> FROM Products p
      -> JOIN Order_Details od
      -> ON p.product_id = od.product_id
      -> GROUP BY p.product_id, p.product_name
      -> ORDER BY Total_Sold DESC
      -> LIMIT 1;
+-----+-----+
| product_name | Total_Sold |
+-----+-----+
| Keyboard    | 13 |
+-----+-----+
1 row in set (0.00 sec)

```

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mysql>
mysql> SELECT d.dept_name,
->      COUNT(e.emp_id) AS Employee_Count
-> FROM Departments d
-> LEFT JOIN Employees e
-> ON d.dept_id = e.dept_id
-> GROUP BY d.dept_id, d.dept_name
-> HAVING COUNT(e.emp_id) = (
->      SELECT MAX(emp_count)
->      FROM (
->          SELECT COUNT(*) AS emp_count
->          FROM Employees
->          WHERE dept_id IS NOT NULL
->          GROUP BY dept_id
->      ) AS x
-> );
+-----+-----+
| dept_name | Employee_Count |
+-----+-----+
| CSE       | 7              |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT DATE_FORMAT(o.order_date, '%Y-%m') AS Month,
->      SUM(od.quantity * p.price) AS Monthly_Sales,
->      SUM(SUM(od.quantity * p.price)
->      OVER (
->          ORDER BY DATE_FORMAT(o.order_date, '%Y-%m')
->      ) AS Running_Sales
-> FROM Orders o
-> JOIN Order_Details od
-> ON o.order_id = od.order_id
-> JOIN Products p
-> ON od.product_id = p.product_id
-> GROUP BY DATE_FORMAT(o.order_date, '%Y-%m')
-> ORDER BY Month;
+-----+-----+-----+
| Month | Monthly_Sales | Running_Sales |
+-----+-----+-----+
| 2026-01 | 191000.00 | 191000.00 |
| 2026-02 | 110000.00 | 301000.00 |
| 2026-03 | 154000.00 | 455000.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT customer_name,
->      total_purchase,
->      RANK() OVER (ORDER BY total_purchase DESC) AS Customer_Rank
```

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C:\Windows\System32\cmd.e X + v
mysql>
mysql> SELECT customer_name,
->         total_purchase,
->         RANK() OVER (ORDER BY total_purchase DESC) AS Customer_Rank
-> FROM (
->   SELECT c.customer_name,
->         SUM(od.quantity * p.price) AS total_purchase
->   FROM Customers c
->   JOIN Orders o
->   ON c.customer_id = o.customer_id
->   JOIN Order_Details od
->   ON o.order_id = od.order_id
->   JOIN Products p
->   ON od.product_id = p.product_id
->   GROUP BY c.customer_id, c.customer_name
-> ) AS x;

```

customer_name	total_purchase	Customer_Rank
Customer1	216000.00	1
Customer2	125000.00	2
Customer4	94000.00	3
Customer3	20000.00	4

4 rows in set (0.00 sec)

```

mysql>
mysql> SELECT e.emp_name AS Employee,
->         e.salary AS Employee_Salary,
->         m.emp_name AS Manager,
->         m.salary AS Manager_Salary
-> FROM Employees e
-> JOIN Employees m
-> ON e.manager_id = m.emp_id
-> WHERE e.salary > m.salary;

```

Employee	Employee_Salary	Manager	Manager_Salary
Vijay	99000.00	Rahul	88000.00
Ravi	104500.00	Rahul	88000.00

2 rows in set (0.00 sec)

```

mysql>
mysql> SELECT email, COUNT(*) AS Count
-> FROM Employees
-> GROUP BY email
-> HAVING COUNT(*) > 1;
Empty set (0.00 sec)

```